

Lecture 1: Regression, identification, and the OLS estimator

Economics 527: Econometric Methods

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Motivation and data

A motivating example: Return to education

- Suppose wages are generated according to:

$$\ln \text{Wage} = \beta_1 + \beta_2 \text{Education} + \beta_3 \text{Gender} + \beta_4 \text{Race} \\ + \beta_5 \text{Urban} + \beta_6 \text{Experience} + U.$$

- We have a sample of workers and, for each worker, we observe Wage, Education, etc.
- U collects everything besides Education, Gender, etc. that affects wages. Ability is first on that list, and it is unobserved (latent).
- The unknown coefficient β_2 measures the return to education, holding everything else fixed.
- Questions/Issues:
 - Why in this equation β_2 measures the return to education;
 - How to “learn” β_2 from random data;
 - What data randomness means;
 - What has to be assumed about the unobserved U to be able to learn β_2 .

Levels and logs

- **Level-level**, $y = \beta x$:

$$\beta = \frac{dy}{dx},$$

and the proportional change in y per unit of x is β/y , which depends on where y is.

- **Log-level**, $\ln y = \beta x$:

$$\beta = \frac{d \ln y}{dx} = \frac{dy/dx}{y},$$

so β is the proportional rate of change of y with respect to x . Multiply by 100 for a percentage.

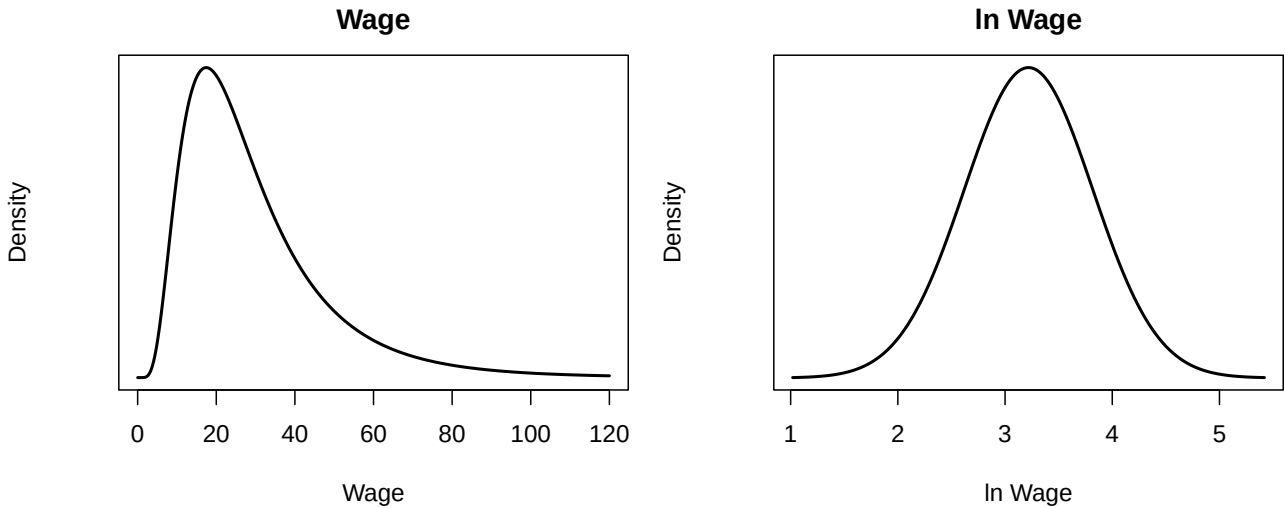
- **Log-log**, $\ln y = \beta \ln x$:

$$\beta = \frac{d \ln y}{d \ln x} = \frac{dy/y}{dx/x},$$

the elasticity of y with respect to x .

Logs often “work better”

- Wages are bounded below by zero and have a long right tail. Their log is close to symmetric.



- A symmetric dependent variable is not required by anything that follows, but specifications with logs often “work better” in practice.

Data

- We observe a **sample** of n workers:

$$\{(Y_i, X_i) : i = 1, \dots, n\}.$$

- For worker i , $Y_i = \ln \text{Wage}_i$ is a scalar (1×1) and X_i stacks the k regressors, with a 1 in the first position for the intercept:

$$X_i = \begin{pmatrix} 1 \\ \text{Educ}_i \\ \text{Gender}_i \\ \vdots \end{pmatrix} \quad (k \times 1), \quad \beta = \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_k \end{pmatrix} \quad (k \times 1).$$

- The wage equation for worker i is

$$Y_i = X_i^\top \beta + U_i = \sum_{j=1}^k X_{ij} \beta_j + U_i,$$

where $X_i^\top \beta = \beta^\top X_i$ is a scalar.

What is random and what is fixed

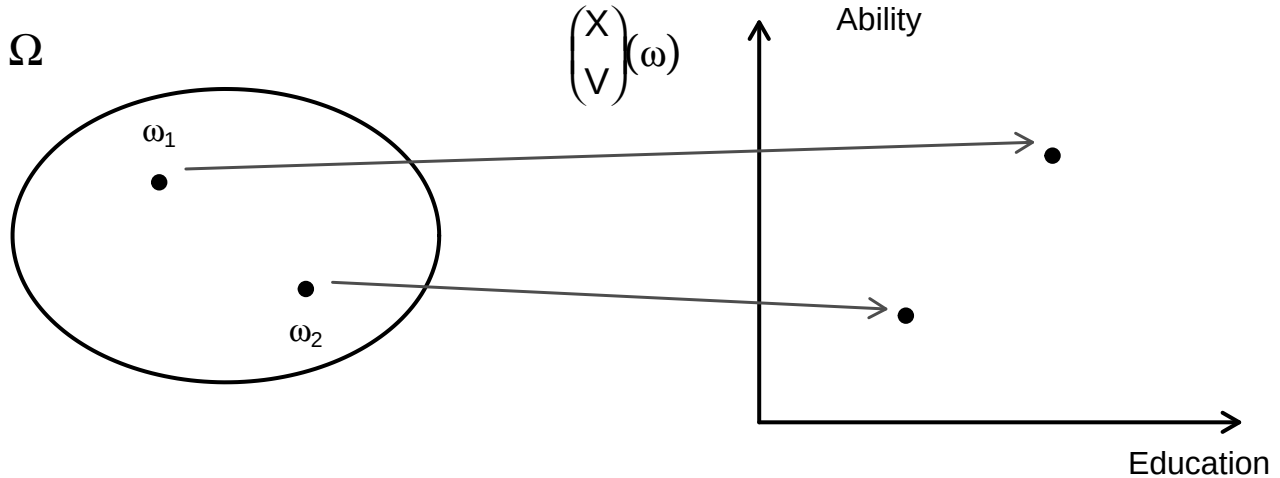
- β is **fixed** and unknown.
- X_i , U_i , and therefore Y_i are **random**.
- The **population** is the distribution that generates $\{(Y_i, X_i) : i = 1, \dots, n\}$, a probability distribution rather than a group of people.
- In a cross-section, each observation is a different worker, so we take the observations to be drawn **independently and identically distributed**:

$$(Y_i, X_i) \text{ are iid, with joint distribution } F_{Y, X_1, \dots, X_k}.$$

- Note that independence is between (Y_i, X_i) and (Y_j, X_j) for $i \neq j$, not between Y_i and X_i .
- Under the iid assumption, we can take the population to be the joint distribution of one observation (Y_i, X_i) .

Modelling the randomness

- A random experiment has a sample space Ω of outcomes ω . These are possible states of the world.
- Let X denote education and V ability.
- The pair (X, V) is a function from Ω to $\mathbb{R}^2$: for the state of the world ω , $X(\omega)$ is the education and $V(\omega)$ is the ability.



- A probability P assigns to each event A a number $P(A) \in [0, 1]$. It induces a distribution for (X, V) .
- The joint CDF of (X, V) is a statement about sets of outcomes: for all $x, v \in \mathbb{R}$,

$$\begin{aligned} F_{X,V}(x, v) &= P(X \leq x, V \leq v) \\ &= P(\{\omega \in \Omega : X(\omega) \leq x, V(\omega) \leq v\}). \end{aligned}$$

Probability

Random variables and the CDF

- A random variable is a map $Y : \Omega \rightarrow \mathbb{R}$. For a set $A \subset \mathbb{R}$,

$$P(Y \in A) = P(\{\omega : Y(\omega) \in A\}).$$

- Taking $A = (-\infty, y]$ gives the **cumulative distribution function**:

$$F_Y(y) = P(Y \leq y), \quad y \in \mathbb{R}.$$

- Properties:
 - F_Y is nondecreasing and right-continuous, $F_Y(y) \rightarrow 0$ as $y \rightarrow -\infty$ and $F_Y(y) \rightarrow 1$ as $y \rightarrow \infty$;
 - $P(a < Y \leq b) = F_Y(b) - F_Y(a)$;
 - a jump of F_Y at y_1 has size $P(Y = y_1)$; where F_Y is continuous, $P(Y = y_1) = 0$.

Continuous random variables and the PDF

- When F_Y is differentiable, the **probability density function** is

$$f_Y(y) = \frac{dF_Y(y)}{dy}, \quad \text{so} \quad F_Y(y) = \int_{-\infty}^y f_Y(v) dv.$$

- $f_Y(y) \geq 0$, $\int_{-\infty}^{\infty} f_Y(y) dy = 1$, and for $A \subset \mathbb{R}$,

$$P(Y \in A) = \int_A f_Y(y) dy.$$

- A density is not a probability: $P(Y = y) = 0$ for every y , and $f_Y(y)$ can exceed one.
- Everything in this course is written for the continuous case. The discrete case replaces integrals with sums.

Expectation

- For a function h ,

$$E[h(Y)] = \int_{-\infty}^{\infty} h(y) f_Y(y) dy.$$

- Throughout, every expectation we write down is assumed to exist and to be finite.
- $E[Y]$ is the first moment, $E[Y^2]$ the second, and

$$\text{Var}(Y) = E[(Y - E[Y])^2] = E[Y^2] - (E[Y])^2.$$

- The mean is the best constant predictor of Y under squared loss:

$$E[(Y - c)^2] = E[(Y - E[Y])^2] + (E[Y] - c)^2 \geq \text{Var}(Y),$$

with equality only at $c = E[Y]$. (The cross term $2(E[Y] - c)E[Y - E[Y]]$ is zero.)

- Under absolute loss the minimiser of $E[|Y - c|]$ is the median instead. Which loss we use decides which feature of the distribution we end up estimating.

Joint distributions

- For two random variables Y and X on the same Ω :

$$F_{Y,X}(y, x) = P(Y \leq y, X \leq x),$$

$$f_{Y,X}(y, x) = \frac{\partial^2 F_{Y,X}(y, x)}{\partial y \partial x}.$$

- $f_{Y,X} \geq 0$, $\iint f_{Y,X}(y, x) dy dx = 1$, and for $A \subset \mathbb{R}^2$,

$$P((Y, X) \in A) = \iint_A f_{Y,X}(y, x) dy dx.$$

- With k regressors the joint CDF of one observation is

$$F_{Y, X_1, \dots, X_k}(y, x_1, \dots, x_k) = P(Y \leq y, X_1 \leq x_1, \dots, X_k \leq x_k).$$

From the joint density to a marginal

- **Claim.** The marginal density of X is obtained by integrating Y out:

$$f_X(x) = \int_{-\infty}^{\infty} f_{Y,X}(y, x) dy.$$

- **Proof.** Start from the marginal CDF and differentiate:

$$\begin{aligned} F_X(x) &= P(X \leq x) = P(X \leq x, Y < \infty) \\ &= \int_{-\infty}^x \int_{-\infty}^{\infty} f_{Y,X}(y, u) dy du, \\ f_X(x) &= \frac{dF_X(x)}{dx} = \frac{d}{dx} \left[\int_{-\infty}^x \int_{-\infty}^{\infty} f_{Y,X}(y, u) dy du \right] \\ &= \int_{-\infty}^{\infty} f_{Y,X}(y, x) dy, \end{aligned}$$

using $\frac{d}{dx} \int_{-\infty}^x h(u) du = h(x)$.

- The converse fails: knowing f_X and f_Y does **not** determine $f_{Y,X}$. The marginals say nothing about how Y and X move together.

Conditional density

- The density of Y **given** $X = x$, defined where $f_X(x) > 0$:

$$f_{Y|X}(y | x) = \frac{f_{Y,X}(y, x)}{f_X(x)}, \quad y \in \mathbb{R}.$$

- This has the same shape as conditional probability for events, $P(A | B) = P(A \cap B)/P(B)$: the joint, rescaled by the marginal density of what we condition on.
- It is a proper density in y :

$$\int_{-\infty}^{\infty} f_{Y|X}(y | x) dy = \frac{1}{f_X(x)} \underbrace{\int_{-\infty}^{\infty} f_{Y,X}(y, x) dy}_{= f_X(x)} = 1.$$

Independence

- Y and X are **independent** if conditioning changes nothing:

$$f_{Y|X}(y | x) = f_Y(y) \quad \text{for all } y \text{ and all } x \text{ with } f_X(x) > 0.$$

- Equivalently the joint factorises:

$$\begin{aligned} f_{Y,X}(y, x) &= f_Y(y) f_X(x) \quad \text{for all } x, y, \\ \text{or equivalently} \quad F_{Y,X}(y, x) &= F_Y(y) F_X(x) \quad \text{for all } x, y. \end{aligned}$$

- Independence is a strong statement: it involves the whole conditional distribution. Regression will need much less.

Expectation of a function of two variables

- For a function h of both arguments,

$$E[h(Y, X)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(y, x) f_{Y,X}(y, x) dy dx.$$

- With $h(y, x) = yx$ this is the joint moment $E[YX]$. Centring both arguments gives the **covariance**:

$$\begin{aligned} \text{Cov}(Y, X) &= E[(Y - E[Y])(X - E[X])] \\ &= E[YX] - E[Y] E[X]. \end{aligned}$$

- When $\text{Var}(Y)$ and $\text{Var}(X)$ are positive and finite, dividing by the two standard deviations removes the units and gives the correlation,

$$\text{Corr}(Y, X) = \frac{\text{Cov}(Y, X)}{\sqrt{\text{Var}(Y) \text{Var}(X)}} \in [-1, 1],$$

where the bound is the Cauchy-Schwarz inequality.

- Y and X are **uncorrelated** if $\text{Cov}(Y, X) = 0$, equivalently $E[YX] = E[Y] E[X]$.
- Uncorrelatedness is a single restriction. It says the best straight-line predictor of Y from X under squared loss has zero slope, and it says nothing about any other way the two variables might be related.

Some implications

- Under independence the expectation of a product of separate functions factorises:

$$\begin{aligned} \mathbb{E}[h_1(Y) h_2(X)] &= \iint h_1(y) h_2(x) \underbrace{f_{Y,X}(y, x)}_{= f_Y(y) f_X(x)} dy dx \\ &= \int h_1(y) f_Y(y) dy \cdot \int h_2(x) f_X(x) dx \\ &= \mathbb{E}[h_1(Y)] \mathbb{E}[h_2(X)]. \end{aligned}$$

- This holds for **every** pair of functions h_1 and h_2 , not just for the identity.
- Taking $h_1(y) = y$ and $h_2(x) = x$ gives $\mathbb{E}[YX] = \mathbb{E}[Y]\mathbb{E}[X]$, so $\text{Cov}(Y, X) = 0$: independence implies uncorrelatedness.
- The converse fails, and it fails for the reason above: uncorrelatedness is the factorisation for one pair of functions, whereas independence is the factorisation for all of them.

Conditional expectation

Conditional expectation: at a point

- Fix x with $f_X(x) > 0$ and take the mean of the conditional density:

$$\mathbb{E}[Y \mid X = x] = \int_{-\infty}^{\infty} y f_{Y|X}(y \mid x) dy.$$

This is a **number**.

- Repeat for every x with $f_X(x) > 0$. The result is a function of x ,

$$x \mapsto \mathbb{E}[Y \mid X = x] =: g(x),$$

the **conditional expectation function**.

Conditional expectation: as a random variable

- Now plug the random variable X into g :

$$\mathbb{E}[Y \mid X] := g(X) = \int_{-\infty}^{\infty} y f_{Y|X}(y \mid X) dy.$$

This is a **random variable**: a different draw of X gives a different value.

- The distinction matters for every proof that follows. $\mathbb{E}[Y \mid X = x]$ is a number indexed by x ; $\mathbb{E}[Y \mid X]$ is the random variable obtained by evaluating that function at X .
- The same construction applied to the squared deviation gives the **conditional variance**,

$$\text{Var}(Y \mid X) = \mathbb{E}[(Y - \mathbb{E}[Y \mid X])^2 \mid X],$$

again a random variable, and a function of X .

- Since $\mathbb{E}[Y \mid X]$ is random, it has a mean of its own. What is it?

Law of iterated expectations

- Claim (LIE).**

$$\mathbb{E}[\mathbb{E}[Y \mid X]] = \mathbb{E}[Y].$$

- Averaging the conditional mean over the distribution of X gives back the unconditional mean.
- Read from right to left, it says any expectation can be computed in two stages: first conditional on X , then over X . Most proofs in this course use it that way.

Law of iterated expectations: proof

- Write the outer expectation as an integral against f_X , then unpack g :

$$\begin{aligned} \mathbb{E}[\mathbb{E}[Y | X]] &= \int_{-\infty}^{\infty} \underbrace{\mathbb{E}[Y | X = x]}_{g(x)} f_X(x) dx \\ &= \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} y f_{Y|X}(y | x) dy \right] f_X(x) dx \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y \underbrace{f_{Y|X}(y | x) f_X(x)}_{= f_{Y,X}(y,x)} dy dx \\ &= \int_{-\infty}^{\infty} y \underbrace{\left[\int_{-\infty}^{\infty} f_{Y,X}(y, x) dx \right]}_{= f_Y(y)} dy \\ &= \int_{-\infty}^{\infty} y f_Y(y) dy = \mathbb{E}[Y]. \end{aligned}$$

- The step from the third line to the fourth changes the order of integration.

Taking out what is known

- Conditional on X , any function of X is a constant and comes out of the expectation:

$$\begin{aligned} \mathbb{E}[Y \cdot h(X) | X] &= \int_{-\infty}^{\infty} y h(X) f_{Y|X}(y | X) dy \\ &= h(X) \int_{-\infty}^{\infty} y f_{Y|X}(y | X) dy = h(X) \mathbb{E}[Y | X]. \end{aligned}$$

- Two special cases used constantly:

$$\mathbb{E}[h(X) | X] = h(X), \quad \mathbb{E}[c | X] = c.$$

Question

- Let Y, X, Z be random variables. Which of the following equals $\mathbb{E}[\mathbb{E}[Y | X, Z] | Z]$?
 - $\mathbb{E}[Y]$
 - $\mathbb{E}[Y | X]$
 - $\mathbb{E}[Y | Z]$
 - 0

Mean independence

- Y is **mean independent** of X if the conditional mean does not move with the conditioning value:

$$\mathbb{E}[Y | X = x] = c \quad \text{for all } x \text{ with } f_X(x) > 0,$$

where the constant c does not depend on x .

- Question.** The constant c does not depend on x . What must it be?

- 0
- $\mathbb{E}[X]$
- $\mathbb{E}[Y]$
- 42

- Mean independence is weaker than independence:

- Under mean independence, $\mathbb{E}[Y | X]$ does not depend on X , but $\mathbb{E}[Y^2 | X]$, $\text{Var}(Y | X)$, etc., can change with X .
- Under independence, the whole conditional distribution is constant in X , so $\text{Var}(Y | X)$, etc., are constant too.

A zero conditional mean

- Suppose

$$E[U | X] = 0.$$

By the previous slide this is mean independence of U from X , in the case where the constant is zero.

- Two consequences, by the LIE and by taking out what is known:

$$E[U] = E \left[\underbrace{E[U | X]}_{=0} \right] = 0,$$

$$E[UX] = E[E[UX | X]] = E \left[X \underbrace{E[U | X]}_{=0} \right] = 0.$$

The second line takes X out of the inner expectation, which is allowed because X is known once we condition on it.

- Putting the two together,

$$\text{Cov}(U, X) = E[UX] - E[U] E[X] = 0 - 0 = 0,$$

so U and X are uncorrelated.

Three concepts of unrelatedness

- Ordered from strongest to weakest:

$$\text{independence} \implies E[U | X] = E[U] \implies \text{Cov}(U, X) = 0,$$

and neither arrow reverses.

- Mean independence allows $\text{Var}(U | X)$ to depend on X ; independence does not. A variable can have a conditional mean that never moves with X while its conditional variance does.
- Uncorrelatedness is weaker still: $\text{Cov}(U, X) = 0$ is a single restriction, and it is compatible with a conditional mean $E[U | X]$ that varies with X , because $\text{Cov}(U, X) = \text{Cov}(E[U | X], X)$.

Conditional mean minimises mean squared error

- Among all functions m of X , the conditional mean is the best predictor of Y under squared loss. Condition on X first:

$$\begin{aligned} E[(Y - m(X))^2 | X] &= E \left[(Y - E[Y | X])^2 | X \right] \\ &\quad + (E[Y | X] - m(X))^2, \end{aligned}$$

because the cross term is $2(E[Y | X] - m(X)) E[Y - E[Y | X] | X] = 0$.

- Now average over X with the LIE:

$$E[(Y - m(X))^2] \geq E \left[(Y - E[Y | X])^2 \right],$$

with equality only when $m(X) = E[Y | X]$ with probability one.

- This is the reason regression is about $E[Y | X]$ and not some other feature of the conditional distribution. Change the loss and a different feature wins.

Linear regression model

Object of interest

- The relationship between the regressors and the dependent variable is summarised by the conditional expectation function:

$$E[Y_i | X_i] = E[Y_i | X_{i1}, \dots, X_{ik}],$$

for instance

$$E[\ln \text{Wage}_i | \text{Education}_i, \text{Experience}_i, \text{Race}_i, \text{Gender}_i, \dots].$$

- The practical problem: with data $\{(Y_i, X_i) : i = 1, \dots, n\}$, estimate $E[Y_i | X_i]$.
- Without restrictions this is a function of k arguments to be learned from n points. The **parametric** approach assumes the functional form and leaves a finite number of unknown constants.

Linear regression model

- Assumption.** The conditional expectation is linear in the parameters:

$$E[Y_i | X_i] = \beta_1 X_{i1} + \dots + \beta_k X_{ik} = X_i^\top \beta,$$

where $X_i = (X_{i1}, \dots, X_{ik})^\top$ and $\beta = (\beta_1, \dots, \beta_k)^\top \in \mathbb{R}^k$.

- Write $E[Y_i | X_i = x] = x^\top \beta = \sum_{l=1}^k \beta_l x_l$. Differentiating in x_j ,

$$\frac{\partial E[Y_i | X_i = x]}{\partial x_j} = \frac{\partial(\beta_1 x_1 + \dots + \beta_k x_k)}{\partial x_j} = \beta_j.$$

- So β_j is the change in the conditional mean of Y_i per unit change in X_{ij} , holding the other regressors fixed: a **marginal effect**.
- “Linear” refers to β . The regressors can be anything: squares, logs, products, indicator variables.

Error term

- Define** the error as the deviation from the conditional mean:

$$U_i := Y_i - E[Y_i | X_i] = Y_i - X_i^\top \beta.$$

- Then, by construction,

$$\begin{aligned} Y_i &= X_i^\top \beta + U_i, \\ E[U_i | X_i] &= \underbrace{E[Y_i | X_i]}_{= X_i^\top \beta} - \underbrace{E[X_i^\top \beta | X_i]}_{= X_i^\top \beta} = 0. \end{aligned}$$

- U_i is not observable: the conditional expectation function is unknown, so the deviation from it is unknown.
- If the pairs (Y_i, X_i) are iid, then so are the U_i , since each is the same function of its own pair.

Two ways to write the same model

- Either start from the conditional mean and define the error,

$$\left. \begin{aligned} E[Y_i | X_i] &= X_i^\top \beta \\ U_i &:= Y_i - X_i^\top \beta \end{aligned} \right\} \implies E[U_i | X_i] = 0,$$

- or start from the equation and the restriction on the error,

$$\left. \begin{aligned} Y_i &= X_i^\top \beta + U_i \\ E[U_i | X_i] &= 0 \end{aligned} \right\} \implies E[Y_i | X_i] = X_i^\top \beta.$$

Intercept

- Consider an example with a single regressor $X_i^* \in \mathbb{R}$.
- Suppose $Y_i = \beta^* X_i^* + U_i^*$ with $E[U_i^* | X_i^*] = \alpha \neq 0$.
- We can add and subtract α to get:

$$Y_i = \alpha + \beta^* X_i^* + \underbrace{(U_i^* - \alpha)}_{U_i}, \quad E[U_i | X_i^*] = 0.$$

- Define vectors $X_i = (1, X_i^*)^\top$ and $\beta = (\alpha, \beta^*)^\top$. Then

$$Y_i = X_i^\top \beta + U_i, \quad E[U_i | X_i] = E[U_i | 1, X_i^*] = 0,$$

as conditioning on 1 in addition to X_i^* does not change the conditional expectation.

- Here α is the intercept: a coefficient on a constant regressor (i.e., a regressor equal to 1 for all observations i).
- We include the intercept to absorb constant but non-zero conditional means of the error. This does not change the slope coefficients.
- However, if the conditional mean of the error is not constant, i.e., $E[U_i^* | X_i^*] = \alpha + \gamma X_i^*$, then

$$Y_i = \alpha + (\beta^* + \gamma) X_i^* + \underbrace{(U_i^* - \alpha - \gamma X_i^*)}_{U_i}, \quad E[U_i | X_i^*] = 0,$$

and the slope coefficient is affected.

Identification

Identification: the question

- Consider the model $Y_i = X_i^\top \beta + U_i$ with $E[U_i | X_i] = 0$:
 - Y_i, X_i are observed,
 - U_i is unobserved,
 - $\beta \in \mathbb{R}^k$ is unknown.
- Suppose the joint distribution of the observables $F_{Y, X_1, \dots, X_k}(y, x_1, \dots, x_k)$ is **known** (we are at the population level). You have no information about U_i beyond $E[U_i | X_i] = 0$.
- Can you recover β ?
- If the answer is no with the whole distribution in hand, no amount of data will help, as the joint distribution of the data does not pin down β .

Identification: the definition

- Definition.** β is **identified** if it is uniquely determined by the joint distribution of the observed variables $(Y_i, X_{i1}, \dots, X_{ik})$.
- Two things can go wrong:
 - the distribution of the observables is consistent with more than one value of β ;
 - it is consistent with none, because the model is wrong.
- We ask: what does $E[U_i | X_i] = 0$ let us compute from the distribution of (Y_i, X_i) , and does it pin β down?

What the model gives us

- By $E[U_i | X_i] = 0$ and the LIE, $E[X_i U_i] = 0$.
- Rewrite U_i in terms of the observables and β :

$$U_i = Y_i - X_i^\top \beta$$

- Obtain the **population normal equations**:

$$0 = E[X_i U_i] = E[X_i(Y_i - X_i^\top \beta)].$$

- Multiply out, one step at a time, distribute the expectation:

$$\begin{aligned} 0 &= E[X_i U_i] = E[X_i(Y_i - X_i^\top \beta)] \\ &= E[X_i Y_i - X_i X_i^\top \beta] \\ &= E[X_i Y_i] - E[X_i X_i^\top \beta] \\ &= E[X_i Y_i] - E[X_i X_i^\top] \beta, \end{aligned}$$

the last step because β is a vector of constants and comes out of the expectation.

- Rearranging,

$$E[X_i X_i^\top] \beta = E[X_i Y_i].$$

- Knowns: $E[X_i Y_i]$ ($k \times 1$) and $E[X_i X_i^\top]$ ($k \times k$), both features of the distribution of the observables. Unknown: β .
- If $E[X_i X_i^\top]$ is invertible, the system has exactly one solution. Whether it is invertible is the whole question.

Second-moment matrix

- Write X_i as a column and $X_i^\top$ as a row, and multiply:

$$\begin{aligned} X_i X_i^\top &= \begin{pmatrix} X_{i1} \\ X_{i2} \\ \vdots \\ X_{ik} \end{pmatrix} (X_{i1} \quad X_{i2} \quad \cdots \quad X_{ik}) \\ &= \begin{pmatrix} X_{i1}^2 & X_{i1}X_{i2} & \cdots & X_{i1}X_{ik} \\ X_{i2}X_{i1} & X_{i2}^2 & \cdots & X_{i2}X_{ik} \\ \vdots & \vdots & \ddots & \vdots \\ X_{ik}X_{i1} & X_{ik}X_{i2} & \cdots & X_{ik}^2 \end{pmatrix} \quad (k \times k). \end{aligned}$$

Entry (j, l) of the product is $X_{ij}X_{il}$.

- The expectation is taken entry by entry:

$$E[X_i X_i^\top] = E \begin{bmatrix} X_{i1}^2 & X_{i1}X_{i2} & \cdots & X_{i1}X_{ik} \\ X_{i2}X_{i1} & X_{i2}^2 & \cdots & X_{i2}X_{ik} \\ \vdots & \vdots & \ddots & \vdots \\ X_{ik}X_{i1} & X_{ik}X_{i2} & \cdots & X_{ik}^2 \end{bmatrix}.$$

- It is **symmetric**: using $(AB)^\top = B^\top A^\top$,

$$(E[X_i X_i^\top])^\top = E[(X_i X_i^\top)^\top] = E[X_i X_i^\top].$$

Second-moment matrix is positive semidefinite

- **Definitions.** A $k \times k$ symmetric matrix A is **positive semidefinite** (p.s.d.) if $c^\top A c \geq 0$ for all $c \in \mathbb{R}^k$, and **positive definite** (p.d.) if $c^\top A c > 0$ for all $c \neq 0$.
- Take any $c \in \mathbb{R}^k$ and define the scalar

$$W_i = X_i^\top c = c_1 X_{i1} + \cdots + c_k X_{ik}, \quad c^\top X_i = (X_i^\top c)^\top = W_i.$$

- Then the quadratic form is the expectation of a square:

$$c^\top \mathbb{E} [X_i X_i^\top] c = \mathbb{E} \left[\underbrace{c^\top X_i}_{\tilde{W}_i} \underbrace{X_i^\top c}_{\tilde{W}_i} \right] = \mathbb{E} \left[\underbrace{W_i^2}_{\geq 0} \right] \geq 0.$$

- So $\mathbb{E} [X_i X_i^\top]$ is p.s.d. whatever the distribution of the regressors. Positive **definiteness** is the extra step, and it can fail.

When positive definiteness fails

- Suppose some $\tilde{c} \neq 0$ gives $\tilde{c}^\top \mathbb{E} [X_i X_i^\top] \tilde{c} = 0$. With $\tilde{W}_i = X_i^\top \tilde{c}$,

$$\begin{aligned} 0 = \tilde{c}^\top \mathbb{E} [X_i X_i^\top] \tilde{c} &= \mathbb{E} [\tilde{W}_i^2] \iff \tilde{W}_i = 0 \text{ with probability } 1 \\ &\iff \boxed{\tilde{c}_1 X_{i1} + \dots + \tilde{c}_k X_{ik} = 0} \end{aligned}$$

- With $\tilde{c} \neq 0$, say $\tilde{c}_1 \neq 0$, one regressor equals, with probability one, an exact linear combination of the others:

$$X_{i1} = -\frac{\tilde{c}_2}{\tilde{c}_1} X_{i2} - \dots - \frac{\tilde{c}_k}{\tilde{c}_1} X_{ik}.$$

This is **perfect multicollinearity**.

Positive definiteness and the rank condition

- Putting the two directions together,

$$\begin{aligned} \mathbb{E} [X_i X_i^\top] \text{ is p.d.} &\iff X_i^\top c = 0 \text{ only if } c = 0 \\ &\iff \mathbb{E} [X_i X_i^\top] c = 0 \text{ only if } c = 0 \\ &\iff \text{rank}(\mathbb{E} [X_i X_i^\top]) = k. \end{aligned}$$

- Ruled out: A relation $c_1 X_{i1} + c_2 X_{i2} = 0$ for some $(c_1, c_2) \neq (0, 0)$.
- $X_{i2} = X_{i1}^2$ is not linear and is fine, unless X_{i1} is binary, when $X_{i1}^2 = X_{i1}$.

Identification: the result

- **Assume**
 1. $Y_i = X_i^\top \beta + U_i$ with $\beta \in \mathbb{R}^k$;
 2. $\mathbb{E} [U_i | X_i] = 0$;
 3. No perfect multicollinearity: $\text{rank}(\mathbb{E} [X_i X_i^\top]) = k$.
- Then β is identified:
 - Assumption 1 and 2 produce: $\mathbb{E} [X_i (Y_i - X_i^\top \beta)] = 0$.
 - This implies $\mathbb{E} [X_i X_i^\top] \beta = \mathbb{E} [X_i Y_i]$.
 - Assumption 3 makes $\mathbb{E} [X_i X_i^\top]$ invertible, so

$$\boxed{\beta = (\mathbb{E} [X_i X_i^\top])^{-1} \mathbb{E} [X_i Y_i]}$$

Estimation and least squares

From identification to an estimator

- Identification wrote β as a function of population moments:

$$\beta = (\mathbb{E} [X_i X_i^\top])^{-1} \mathbb{E} [X_i Y_i].$$

- **Method of moments.** Replace each expectation $E[\cdot]$ by the sample average $\frac{1}{n} \sum_{i=1}^n (\cdot)$:

$$\hat{\beta} = \left(\frac{1}{n} \sum_{i=1}^n X_i X_i^\top \right)^{-1} \frac{1}{n} \sum_{i=1}^n X_i Y_i = \left(\sum_{i=1}^n X_i X_i^\top \right)^{-1} \sum_{i=1}^n X_i Y_i.$$

The two factors of $\frac{1}{n}$ cancel.

- An **estimator** is any function of the sample $\{(Y_i, X_i)\}$. It may not depend on U_i or on β , which are unobserved.

Same estimator from the moment condition

- Alternatively, start from $E[X_i(Y_i - X_i^\top \beta)] = 0$ and replace the expectation by an average, evaluated at the estimator:

$$\frac{1}{n} \sum_{i=1}^n X_i (\underbrace{Y_i - X_i^\top \hat{\beta}}_{=: \hat{U}_i}) = 0 \quad \Rightarrow \quad \sum_{i=1}^n X_i X_i^\top \hat{\beta} = \sum_{i=1}^n X_i Y_i,$$

which is the same $\hat{\beta}$.

- $\hat{U}_i = Y_i - X_i^\top \hat{\beta}$ is the **residual**: the sample counterpart of the error $U_i = Y_i - X_i^\top \beta$. Unlike U_i , it is computable.
- $\hat{\beta}$ is built to make the residuals satisfy

$$\frac{1}{n} \sum_{i=1}^n X_i \hat{U}_i = 0,$$

which is the sample version of $E[X_i U_i] = 0$.

Estimator is not the parameter

- Sample averages are random; expectations are fixed numbers. In general

$$\frac{1}{n} \sum_{i=1}^n X_i X_i^\top \neq E[X_i X_i^\top], \quad \frac{1}{n} \sum_{i=1}^n X_i Y_i \neq E[X_i Y_i],$$

and so in general $\hat{\beta} \neq \beta$.

- Flip a fair coin ten times: the sample proportion of heads is rarely one half, although the population proportion is exactly one half.
- If the errors $U_1, \dots, U_n$ are continuously distributed, $P(\hat{\beta} = \beta) = 0$.
- $\hat{\beta}$ is a random vector: a different sample gives a different value. It has a distribution, and the next lecture is about where that distribution sits relative to β and how spread out it is.

Sample orthogonality

- Since $\hat{\beta}$ solves $\sum_i X_i (Y_i - X_i^\top \hat{\beta}) = 0$,

$$\sum_{i=1}^n X_i \hat{U}_i = 0 \quad \Leftrightarrow \quad \begin{pmatrix} \sum_{i=1}^n X_{i1} \hat{U}_i \\ \vdots \\ \sum_{i=1}^n X_{ik} \hat{U}_i \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}.$$

- If the first regressor is the constant, $X_{i1} = 1$, the first equation says $\sum_{i=1}^n \hat{U}_i = 0$: the residuals average to zero.

Matrix notation

- Stack the n observations:

$$Y = \begin{pmatrix} Y_1 \\ \vdots \\ Y_n \end{pmatrix}_{n \times 1}, \quad U = \begin{pmatrix} U_1 \\ \vdots \\ U_n \end{pmatrix}_{n \times 1}, \quad X = \begin{pmatrix} X_1^\top \\ \vdots \\ X_n^\top \end{pmatrix}_{n \times k}.$$

- Row i of X is the transpose of the regressor vector X_i ; column j of X holds regressor j for all n observations:

$$X = \begin{pmatrix} X_{11} & X_{12} & \cdots & X_{1k} \\ \vdots & \vdots & & \vdots \\ X_{n1} & X_{n2} & \cdots & X_{nk} \end{pmatrix}.$$

- The n equations $Y_i = X_i^\top \beta + U_i$ become one:

$$Y = \begin{pmatrix} X_1^\top \beta \\ \vdots \\ X_n^\top \beta \end{pmatrix} + \begin{pmatrix} U_1 \\ \vdots \\ U_n \end{pmatrix} \iff Y = \underset{n \times k}{X} \underset{k \times 1}{\beta} + \underset{n \times 1}{U}.$$

Two identities and the matrix form of the estimator

- Claim.**

$$X^\top Y = \sum_{i=1}^n X_i Y_i \quad (k \times 1), \quad X^\top X = \sum_{i=1}^n X_i X_i^\top \quad (k \times k).$$

- Proof of the second.** Transposing the stack of rows gives a row of columns:

$$\begin{aligned} X^\top X &= \begin{pmatrix} X_1^\top \\ \vdots \\ X_n^\top \end{pmatrix}^\top \begin{pmatrix} X_1^\top \\ \vdots \\ X_n^\top \end{pmatrix} = (X_1 \quad \cdots \quad X_n) \begin{pmatrix} X_1^\top \\ \vdots \\ X_n^\top \end{pmatrix} \\ &= \underbrace{X_1 X_1^\top}_{k \times k} + \cdots + X_n X_n^\top = \sum_{i=1}^n X_i X_i^\top. \end{aligned}$$

The first identity is the same computation with the second factor X replaced by Y , whose i th row is the scalar Y_i .

- Substitute the two identities into the method-of-moments formula:

$$\begin{aligned} \hat{\beta}_{k \times 1} &= \left(\sum_{i=1}^n X_i X_i^\top \right)^{-1} \sum_{i=1}^n X_i Y_i \\ &= \left(X^\top X \right)^{-1}_{k \times k} X^\top Y_{k \times 1}. \end{aligned}$$

Residuals in matrix form

- Residual vector and its defining property:

$$\begin{aligned} \hat{U} &= Y - X \hat{\beta}, \\ X^\top \hat{U} &= X^\top (Y - X \hat{\beta}) = X^\top Y - X^\top X \underbrace{(X^\top X)^{-1} X^\top Y}_{\hat{\beta}} \\ &= X^\top Y - X^\top Y = 0. \end{aligned}$$

Sample rank condition

- $\text{rank}(X^\top X) = \text{rank}(X)$.
- Since $X^\top X$ is $k \times k$, its inverse exists exactly when $\text{rank}(X) = k$, meaning the columns of X are linearly independent.
- Let $c \in \mathbb{R}^k$ and consider the linear combination of the columns of X :

$$Xc = c_1 \begin{pmatrix} X_{11} \\ \vdots \\ X_{n1} \end{pmatrix} + \dots + c_k \begin{pmatrix} X_{1k} \\ \vdots \\ X_{nk} \end{pmatrix} = 0 \quad \text{only if } c = 0,$$

In particular $n \geq k$: at least as many observations as coefficients.

- What it means observation by observation: $c_1 X_{i1} + \dots + c_k X_{ik} = 0$ for all $i = 1, \dots, n$ only if $c = 0$.
- But this is equivalent to the population condition $\text{rank}(\mathbb{E}[X_i X_i^\top]) = k$, when $n \geq k$.

Least squares

- Take a candidate value $b \in \mathbb{R}^k$ and measure how badly the fitted values $X_i^\top b$ fit the data by the **sum of squared residuals**:

$$\begin{aligned} S(b) &= \sum_{i=1}^n (Y_i - X_i^\top b)^2 = \begin{pmatrix} Y_1 - X_1^\top b \\ \vdots \\ Y_n - X_n^\top b \end{pmatrix}^\top \begin{pmatrix} Y_1 - X_1^\top b \\ \vdots \\ Y_n - X_n^\top b \end{pmatrix} \\ &= (Y - Xb)^\top (Y - Xb) = \|Y - Xb\|^2, \end{aligned}$$

where for $a \in \mathbb{R}^n$, $\|a\|^2 = a^\top a = \sum_{i=1}^n a_i^2$.

- The **ordinary least squares** estimator picks the b that fits best:

$$\hat{\beta}^{\text{OLS}} = \arg \min_{b \in \mathbb{R}^k} \sum_{i=1}^n (Y_i - X_i^\top b)^2 = \arg \min_{b \in \mathbb{R}^k} \|Y - Xb\|^2.$$

- **Claim.** $\hat{\beta}^{\text{OLS}} = \hat{\beta} = (X^\top X)^{-1} X^\top Y$: least squares and the method of moments deliver the same estimator.

Proof

- We will use:

- $(u + v)^\top (u + v) = u^\top u + v^\top v + 2u^\top v$.
- Add and subtract $X\hat{\beta}$ inside the criterion:

$$Y - Xb = Y - X\hat{\beta} + X\hat{\beta} - Xb = \underbrace{Y - X\hat{\beta}}_{\hat{U}} + X(\hat{\beta} - b).$$

- We have:

$$\begin{aligned} (Y - Xb)^\top (Y - Xb) &= \hat{U}^\top \hat{U} + (\hat{\beta} - b)^\top X^\top X(\hat{\beta} - b) \\ &\quad + 2\hat{U}^\top X(\hat{\beta} - b) \end{aligned}$$

- The cross term vanishes by the normal equations $X^\top \hat{U} = 0$:

$$\boxed{(Y - Xb)^\top (Y - Xb) = \underbrace{\hat{U}^\top \hat{U}}_{\text{free of } b} + (\hat{\beta} - b)^\top X^\top X(\hat{\beta} - b).}$$

- So minimising $(Y - Xb)^\top (Y - Xb)$ is the same as minimising $(\hat{\beta} - b)^\top X^\top X(\hat{\beta} - b)$.

- $(\hat{\beta} - b)^\top X^\top X(\hat{\beta} - b) \geq 0$ as this is a quadratic form in $X^\top X$, which is always p.s.d.
- When $\text{rank}(X) = k$, $X^\top X$ is p.d. and

$$(\hat{\beta} - b)^\top X^\top X(\hat{\beta} - b) = 0 \iff \hat{\beta} - b = 0.$$

- Hence, $\hat{\beta}$ is the unique minimiser of $(Y - Xb)^\top(Y - Xb)$.

Other linear estimators, for comparison

- Other estimators can be easily constructed.
- For a known symmetric positive definite $n \times n$ matrix W , let

$$\tilde{\beta}_W = (X^\top W X)^{-1} X^\top W Y$$

- Note $W = I_n$ gives back $\hat{\beta}$.
- With W diagonal, $W = \text{diag}(w_1, \dots, w_n)$ and $w_i > 0$, this is **weighted least squares**:

$$\tilde{\beta}_W = \left(\sum_{i=1}^n w_i X_i X_i^\top \right)^{-1} \sum_{i=1}^n w_i X_i Y_i.$$

- Which of these is a good estimator, and in what sense, is the subject of the next lecture: bias, variance, and the Gauss-Markov theorem.

Summary

Summary: the model and identification

- The object of interest is the conditional expectation $E[Y_i | X_i]$; the linear model assumes $E[Y_i | X_i] = X_i^\top \beta$, which is the same as $Y_i = X_i^\top \beta + U_i$ with $E[U_i | X_i] = 0$.
- Probability supplied what was needed on the way: densities and conditional densities, the law of iterated expectations, taking out what is known, and mean independence.
- **Identification.** $E[U_i | X_i] = 0$ gives $E[X_i U_i] = 0$, hence the population normal equations $E[X_i(Y_i - X_i^\top \beta)] = 0$, that is $E[X_i X_i^\top] \beta = E[X_i Y_i]$. If $E[X_i X_i^\top]$ is positive definite, which means no exact linear relation among the regressors, then

$$\beta = (E[X_i X_i^\top])^{-1} E[X_i Y_i].$$

Summary: the estimator

- **Estimation.** Under the sample rank condition $\text{rank}(X) = k$, replacing expectations by sample averages gives

$$\hat{\beta} = \left(\sum_{i=1}^n X_i X_i^\top \right)^{-1} \sum_{i=1}^n X_i Y_i = (X^\top X)^{-1} X^\top Y,$$

which is also the unique minimiser of $\sum_i (Y_i - X_i^\top \beta)^2$: the OLS estimator.

- $\hat{\beta}$ solves the sample normal equations $X^\top(Y - X\hat{\beta}) = 0$, that is $X^\top \hat{U} = 0$: the sample counterpart of the population normal equations $E[X_i(Y_i - X_i^\top \beta)] = 0$.
- $\hat{\beta}$ is also the OLS estimator as it solves $\min_b \|Y - Xb\|^2$.